SOFTWARE REQUIREMENTS SPECIFICATION

RealmForge

Team 20
Aeolus Software

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Revision History

Name	Date	Reason For Changes	Version
Andy Huynh	2026-10-05	First Draft	1.0
Andreas Sideris	2026-10-10	P3 and P4 drafts	1.0

This document follows the requirements documentation structure presented in the Handbook of requirements and business analysis, by Bertrand Meyer.

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Goals

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Comment: *Goals are "needs of the target organization, which the system will address". While the development team is the principal user of the other books, the Goals book addresses a wider audience: essentially, all stakeholders.*

G.1 Context and overall objective

Comment: *High-level view of the project: organizational context and reason for building a system.*
Comment: *This chapter should not be empty!*

Goal 1. This is a goal example. If you need explicit (and automatic) numbering, you can use the definitions in the `.tex` template. Is is refined by 3

Requirement 2. *This is a requirement example. It illustrates how numbering is continuous and cross-types (if this is what you need).*

G.2 Current situation

Comment: *Current state of processes to be addressed by the project and the resulting system.*

Requirement 3. *This is a requirement example. It refines 1*

G.3 Expected benefits

Comment: *New processes, or improvement to existing processes, made possible by the projectâ€™s results.* **Comment:** *This chapter should not be empty!*

G.4 Functionality overview

Comment: *Overview of the functions (behavior) of the system. Principal properties only (details are in the System book).*

Setting properties: The system application starts up and the the user is shown options that will allow the user to pick and change various settings during the start of the application, so that the user can experiment with different variants of the game. Comment: *Less detail on the button pressing*

Traversal: After the player has changed some properties or just simply pressed the play option menu button, the game begins and the system displays the HUD as well other UI components such as the main game screen and travel options. The player can navigate to another area by pressing the up, down, left, or right menu option buttons go to another area indicated by the change in the background scenery.

Comment: *Branching patchs, endpoints, you need to win*

Combat: The player enters combat randomly when traversing through areas. Once in combat, the battle screen is shown as well as the enemy the player has to defeat. The player is shown options to attack or escape during the encounter. If the player chooses to attack the enemy, an attack action is displayed followed by the enemies opposing attack. If the player defeats the enemy, the player is rewarded with resources at the end. Comment: *You win or die*

Returning to the Hub: If the player escapes, then there is no reward and the player goes back to the main game screen. The gameplay loop of traveling and encounter enemies continues until the player returns to the starting zone (the main hub), with the return option or when the player is defeated. If the player returns to the starting zone, the player is allowed to spend the resources accumulated throughout the journey to construct buildings and upgrades that will enhance the experience of the next gameplay loop. Comment: *Change to only comming back when you die* Comment: *Immediately enter combat* Comment: *Remove randomization* Comment: *You know what your getting into, but not complete knowledge* Comment: *Higher risk, higher reward*

End of game: Once the player finishes spending resources, the player may return to the main gameplay loop of traveling and fighting enemies to gain resources. If the player is defeated, the game ends and player is returned to the start of the game where the player chooses options for changing parameters.

Non-Functional Requirements

G.5 High-level usage scenarios

Comment: *Fundamental usage paths through the system.*

G.6 Limitations and exclusions

Comment: *Aspects that the system need not address.*

G.7 Stakeholders and requirements sources

Comment: *Groups of people who can affect the project or be affected by it, and other places to consider for information about the project and system.* **Comment:** *This chapter should not be empty!*

Environment

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Comment: *The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate.*

E.1 Glossary

Comment: *Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. This chapter should not be empty!*

- **User** - The main stakeholder that is operating the system; providing inputs and receiving outputs.
- **Player** - Is another name refering the user.

- **User-Interface (UI)** - The entire set of displayed service elements the player used to navigate the system.
- **Heads-Up-Display (HUD)** - The set of interface elements displayed ontop of the actually gameplay that contains vital information helpful to the player or user.

E.2 Components

Comment: *List of elements of the environment that may affect or be affected by the system and project. Includes other systems to which the system must be interfaced.*

E.3 Constraints

Comment: *Obligations and limits imposed on the project and system by the environment.* **Comment:** *This chapter should not be empty!*

E.4 Assumptions

Comment: *Properties of the environment that may be assumed, with the goal of facilitating the project and simplifying the system.*

E.5 Effects

Comment: *Elements and properties of the environment that the system will affect.*

E.6 Invariants

Comment: *Properties of the environment that the systemâ€™s operation must preserve.*

System

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S.3 Interfaces 11
S.4 Detailed usage scenarios 11
S.5 Prioritization 11
S.6 Verification and acceptance criteria 11

Comment: *The System book refines the Goal one by focusing on more detailed requirements about the system under development, mainly its constituents, behaviors and properties.*

S.1 Components

Comment: *Overall structure expressed by the list of major software and, if applicable, hardware parts.*

Comment: *This chapter should not be empty!*

S.2 Functionality

Comment: *One section, S.2.n, for each of the components identified in S.2, describing the corresponding behaviors (functional and non-functional properties).* **Comment:** *This chapter should not be empty!*

S.3 Interfaces

Comment: *How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs).*

S.4 Detailed usage scenarios

Comment: *Examples of interaction between the environment (or human users) and the system: use cases, user stories.*

S.5 Prioritization

Comment: *Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality.*

S.6 Verification and acceptance criteria

Comment: *Specification of the conditions under which an implementation will be deemed satisfactory.*

Project

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Comment: *The Project book describes all the constraints and expectations not about the system itself, but about how to develop and produce it.*

P.1 Roles and personnel

Comment: *Main responsibilities in the project; required project staff and their needed qualifications.*

P.2 Imposed technical choices

Comment: *Any a priori choices binding the project to specific tools, hardware, languages or other technical parameters.*

P.3 Schedule and milestones

The project's primary schedule will follow the milestones and dates below, each named according to the deliverables they must produce (as discussed in the project's development plan):

- M1 Problem Statement, Development Plan, PoC Plan, and Team Charter (Due by September 28, 2026)
- M2 SRS and Hazard Analysis Revision 0 (Due by October 19, 2026)
- M3 VnV Plan Revision 0 (Due by November 2, 2026)
- M4 Design Document Revision -1 and CD-Fall (FODA) (Due by November 16, 2026)
- M5 PoC Demo (Due by November 23, 2026)
- M6 Design Documentation Revision 0 (Due by January 18, 2027)
- M7 Revision 0 Demo (Due by February 1, 2027)
- M8 VnV Report Revision 0 and CD-Winter Revision 0 (Usability Report) (Due by March 8, 2027)

- M9 Final Demonstration Revision 1, EXPO Demonstration (Due by March 15, 2027)
- M10 Final Documentation (Revision 1) (Due by March 31, 2027)

P.4 Tasks and deliverables

The tasks required to be completed to develop RealmForge (and its feature variation capabilities) will be managed using an iterative and incremental approach. Within each milestone (in this context the milestones associated with source code development), team members will divide the work to be done into suitable issues, viewable in RealmForge's GitHub Project. The issues will be coordinated using a Kanban method, tracking their progress and review status. The software features associated with these issues will be regularly planned, developed, reviewed, tested, and deployed using a feature branch workflow, contributing towards each set milestone.

A schedule of development deliverables pertaining to the milestones listed above is available below,

complete with relevant requirement sets that are planned to be implemented at the completion of each deliverable. Milestones that purely involve documentation development are omitted, as this schedule is intended to outline the completion of software development tasks and their associated requirements.

Detailed Schedule:

- M5 PoC Demo
  - **Deadline:** November 23, 2026
  - **Software Deliverables:** RealmForge Proof of Concept Build
  - **High Priority Requirements:** NEED TO ADD
  - **Requirement Sets Implemented:** NEED TO ADD
- M7 Revision 0 Demo
  - **Deadline:** February 1, 2027
  - **Software Deliverables:** RealmForge Revision 0 Build

  - **High Priority Requirements:** NEED TO ADD
  - **Requirement Sets Implemented:** NEED TO ADD
- M9 Final Demonstration Revision 1, EXPO Demonstration
  - **Deadline:** March 15, 2027
  - **Software Deliverables:** RealmForge Revision 1 Build
  - **High Priority Requirements:** NEED TO ADD
  - **Requirement Sets Implemented:** NEED TO ADD

P.5 Required technology elements

Comment: *External systems, hardware and software, expected to be necessary for building the system.*

P.6 Risks and mitigation analysis

Comment: *Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise.*

P.7 Requirements process and report

Comment: *Initially, description of what the requirements process will be; later, report on its steps.*